

POWERBOX · DESIGN DECISIONS REGISTER

PowerBox design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	PBX-DEC-001	Version	0.1
Status	Draft	Date	2026-10-01
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/powerbox		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; budget treated as a value-engineering target

PowerBox design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design for construction (lid over the walls, runners with guide lips, folded shelf and brackets, door with hinge, cam latch and hasp, panel plates, protection plate, corner tabs, filter frame, 76 mm fan hole, feet)	(a) accept P1 to P12; (b) accept with changes	(a)	The whole build plan	PBX-DDR-003, A4
2	Door hinge edge	(a) front edge, as modelled; (b) back edge, as in the product renders, which needs the opening narrowed or the case made deeper	(a): only the front edge has room for a hinge	Door, hinge and step 12	PBX-DDR-003, A1
3	Theft resistance of the pack door	(a) thumb-turn cam latch plus padlock hasp, as modelled; (b) keyed cam latch, about USD 3 more; (c) a locking bay	(a) for the prototype; decide with the first partner	Door latch	PBX-DDR-001, item 15; PBX-DDR-003, A2
4	R10 mass margin of 0.58 kg on estimated masses	(a) accept and weigh at TRL 4; (b) save mass now (1.5 mm panel plates, lighter runner infill)	(a)	None now	PBX-DDR-003, A3
5	R2 shortfall: 1.87 evenings against 2	(a) relax R2 to 1.87 evenings; (b) discharge to 3.5 % state of charge; (c) revise the reference evening with users (about 189 Wh at the loads)	None made at TRL 2	Host firmware cut-off only; no hardware change	PBX-DDR-001, item 13
6	First co-design partner	An NGO running community charging points, a solar home system distributor, or a university group in an outage-prone city	None; partners are chosen per area later	AC socket type, load profile, door lock	PBX-DDR-001, item 14
7	Energy metering on the display for charging points that charge per phone	(a) none; (b) per-port energy count in host firmware; (c) a separate meter	None yet	Host firmware only	PBX-DDR-001, item 16
8	Can a SwapCell pack in legacy discharge accept a station heartbeat and move to mode 2 or 4 without opening its output?	For SwapCell to answer	Raised with SwapCell	Host firmware; first power-up check	PBX-DDR-001, item 18

#	Decision needed	Options	Recommendation	Affects in the build	Source
	PowerBox's host is powered from the pack				
9	Case finish	(a) bare 5052 aluminium, as in the BOM; (b) powder coat, as in the product renders	(b), priced at TRL 4	Body, lid and door finish	REVIEW 2026-09-26, item 4
10	AC socket type	Chosen with the first partner's country	Decide with item 6	AC outlet cut-out in the output panel plate	REVIEW 2026-09-26, item 5
11	Clear window in the pack bay door, as in the product renders	(a) no window, as modelled; (b) a polycarbonate window	(a) until TRL 4	Door	REVIEW 2026-09-26, item 2
12	Lit main switch, as in the product renders	(a) lit rocker switch; (b) plain switch, as in the BOM	(a)	Switch cut-out	REVIEW 2026-09-26, item 6

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The inverter's output is isolated from its DC input so its neutral can be bonded to the case and earth pin; its idle draw and light-load efficiency	The 30 mA RCD only works with a bonded neutral; idle draw sets the auto-off case	PBX-CAL-001 section 8; PBX-PRC-001
2	The charge controller reaches 90 % or more at 12 V and 200 W and at 40 W, and tracks stably	R6 is at risk on the loss model	PBX-CAL-001 section 3
3	The fuse block and its blade fuses are rated 58 V DC or more, and every 48 V fuse 60 V DC or more with at least 1 kA breaking capacity	A short at the pack could draw about 496 A	PBX-CAL-001 section 6; PBX-DDR-003
4	The hole patterns of the class D catch, the receptacle's floating mount, and the inverter's and converter's feet	The catch bracket web, receptacle bracket web and floor holes are drilled to suit	PBX-DDR-003
5	The cam latch grips a 1.2 mm door against a 1.2 mm wall, and its tongue reaches 15 mm	The tongue must turn behind the end wall	PBX-DDR-003, P7
6	The cut-out sizes of every module in the output and input panel plates	The plate cut-outs are drawn from envelopes, not datasheets	PBX-DWG-110, PBX-DWG-111
7	The rivet nut hole size (6.0 mm assumed for M4) and the heat-set insert hole sizes (5.6 mm for M4, 4.0 mm for M3)	Each maker gives its own	PBX-DDR-003
8	The fan's flow through the filter, grille and 76 mm hole (half of 16 L/s free air assumed)	The 6 K air rise at full load depends on it	PBX-CAL-001 section 9

■ Value engineering

Value-engineering target: USD 450 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 482 for the PowerBox parts, excluding the shared SwapCell pack (USD 32 over the target). Main cost drivers and savings worth trying:

- The largest lines are the inverter (USD 70), the grid charger brick (USD 50), the output panel modules (USD 46), the protection parts and wiring (USD 38), the enclosure body (USD 38), the pack bay (USD 40), and the charge controller and buck converter (USD 30 each).
- Making the design constructable added USD 33 over the concept's USD 449: rivet nuts and windows in the body, the deeper lid skirt, the handle doubler, the shelf, brackets, runners and rail, the door hinge, cam latch and staple, the filter frame, the protection plate and the counted fixings.
- Savings worth trying: leave out the grid charger where a SwapCell dock is on hand (USD 50, bringing the parts to USD 432); the DC-only variant without the inverter and AC outlet (about USD 92, and no mains voltage in the box); quotes for the folded sheet parts as one batch from one shop; printing the brackets in place of folding them is not worth trying, since they carry the pack's latch load.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-24	StepGen removed as a charging source; the DC input stays for solar and other DC sources	Amish, 2026-09-24	PBX- PRC-001 v0.3
2026-09-25	TRL 2 items 1 to 12: removable SwapCell pack, side-loading bay, one DC input with MPPT, grid charging through a certified brick only, 300 W AC outlet with a DC-only variant, folded aluminium case, 230 V 50 Hz with a 30 mA RCD, budget for the PowerBox parts only, no human-powered input, station mode, wake by coded INTERLOCK, class D catch	Amish: go with recommendation	PBX- DDR-001
2026-09-25	Recessed normally closed wake button, 30 A buck converter, fuse ratings of PBX-CAL-001	Amish: "i accept all your recommendations, go with them across all repos."	PBX- DDR-002
2026-09-30	Fix the design assumptions so the design can be built, and record the changes	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	PBX- DDR-003 (Draft, open for review)
2026-10-01	The budget is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens."	This register; PBX- CAL-001 v0.2